Lasers Med Sci
DOI 10.1007/s10103-014-1574-8

ORIGINAL ARTICLE

Effects of diode laser on direct pulp capping treatment

A pilot study

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Received: 1 November 2013 / Accepted: 30 March 2014
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Abstract At the present time, evidence-based, best practices have yet to be established for maintaining the vitality of teeth by managing caries-associated pulp exposure in permanent teeth. In terms of biomechanical and esthetic considerations, pulp capping has proven to be more effective than root-canal therapy. Given the low success rate of conventional methods, new techniques, such as laser-assisted repairs, should be developed. The purpose of this study was to compare the effectiveness of conventional and diode laser-assisted methods in direct pulp capping of carious teeth. Ten patients ranging in age from 12 to 40 were assigned randomly to experimental and control groups in this clinical trial in which the participants' teeth were treated with different techniques, i.e., conventional treatment and diode 808-nm, laser-assisted treatment. For each of these groups, five cases were chosen for treatment with the same method under rubber dam isolation. The data were analyzed by the runs test using SPSS software. The success rate was significantly different between conventional (60 %) and diode 808-nm, laser-assisted (100 %) groups after one year ($P>0.05$). The laser-assisted procedure proved to be more effective than the conventional technique in enhancing the outcomes of pulp-capping therapy in carious exposures.

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Keywords Dental pulp capping · Diode laser · Tertiary dentin · Glass ionomer

Introduction

Background

Pulp capping was first performed in 1765 by Philip Pfaff using gold foil. In 1923, Davis suggested using a complex of zinc sulfate and calcium sulfate with zinc oxide for direct pulp capping. Calcium hydroxide was used for pulp capping for the first time by Hermann (1930), and, from the early 1940s to the present time, it has been the material of choice for pulp capping [1–4]. Pulp capping therapy is based on the healing capacity of pulp tissue [5], and it marks the boundary between conservative and aggressive endodontic treatments with great controversy among different authors.

Reviewing previous studies showed high success rates for traumatic pulp exposure [6], whereas the success rate in treating carious exposure with bacterial contamination has been low [7]. Dental specialists use different approaches to treat exposed pulp, including pulp capping, pulpotomy, and root canal therapy. Some authors consider pulp capping as the first choice of treatment, and, depending on the outcome, it may be followed by endodontic treatment [8, 9].

The most visible response to the repair of a pulp injury is the formation of the tertiary dentin matrix. Unlike primary and secondary dentin that forms along the entire pulp–dentin interface, tertiary dentin is secreted by odontoblasts, strictly targeting the affected area. The process of secreting tertiary dentin may be classified as being of reactionary or reparative origin, according to the severity of the initiated response and the conditions in which the new tissue matrix is formed. Usually, reactionary dentin is secreted by existing odontoblasts, and reparatory dentin is secreted by newly

Published online: 23 April 2014

differentiated, odontoblast-like cells. Many factors influence secretion of tertiary dentin, as, degree of pulp damage, existence of debris at the exposure environment, the properties of capping material, pre-existence of inflammation, and bacterial contamination [10].

Statement of the problem

Present endodontic techniques substitute the vital pulp tissue with synthetic materials instead of living tissue. Overobturated materials can produce a foreign body reaction [11] and lose the ability to sense environmental changes of non-vital teeth, making the progression of caries imperceptible by patients. Vitality preservation of the dental pulp tissue is that doing so also maintains the capacity for the partial regeneration of dentin. The formation of reparative dentin is especially important for immature young teeth, since of the partial progress of their apical region and dentinal walls. The structural integrity of teeth that have been subjected to endodontic treatment also may be undermined if they are not properly restored, because it makes them more vulnerable to masticatory forces [12]. In terms of esthetics, root canal therapy can every so often cause in discoloration of the crown of the tooth, mainly due to staining from the endodontic obturation material [13]. Preservation of the pulp vitality in addition helps reduce the incidence of apical periodontitis by bacterial seal infections [14, 15]. The vitality of the pulp tissue is critical, as it gives nutrition and sense to notice pathogen microorganism intoxication [16]. Based on these issues and concerns and given the high cost of endodontic treatments, the ability to maintain the vitality of dental pulp would be preferable to present root canal therapy [17].

Objectives

A review of the pertinent literature indicated that success in both procedures, i.e., conventional and laser-supported procedures, basically depends on the degree to which the wound area can be maintained free of bacteria in the short-term and in the long-term. If this is accomplished effectively, the end result should be the formation of a complete bridge of reparative dentin in direct contact with nearby reactionary dentin as long as the tooth has the appropriate physical and mechanical properties. The aim of this study was to investigate the multiple capability of the diode 808-nm, laser-assisted approach to improve the outcomes of direct pulp capping and compare its outcomes with those of the conventional technique for permanent teeth using the same capping material. In this study, we attempted to determine the relationship between the diode 808-nm, laser-assisted support of direct pulp capping of carious exposures and the results achieved by the conventional technique that has a low success rate in preventing bacterial contamination.

Materials and methods

Research design and subjects

Given that this was a pilot study, first, we evaluated ten patients who had undergone conservative treatment for primary deep caries in their permanent teeth. The patients ranged in age from 12 to 40, and the average age was 26 [Table 1]. Ten teeth from the ten patients who met the criteria of the study were assigned randomly to two different groups.

Speaking about limitation, whereas there was not any study about the use of diode laser in direct pulp capping, it confronted unavailability of sources, another limitation, although the number of cases was considered small but clinical research on patients are difficult to control and collect. Also finding patients with the research set criteria is very difficult and require long time.

Selection criteria

The following selection criteria were used in the research:

Inclusion criteria

1. Permanent teeth with deep caries
2. Vitality of teeth
3. No periapical radiographic changes (Fig. 1)
4. Acceptance of the research program

Exclusion criteria

1. Symptoms
2. Periapical radiographic changes
3. Bleeding continued for more than 3 min after exposure
4. Lack of acceptance of the study's guidelines

Ethical consideration and trial registration

This research was approved by Aachen dental laser center (AALZ), Rheinisch-Westfaelische Technische Hochschule (RWTH) University. After receiving a thorough explanation of the goals of the experiment, clinical procedures, and possible risks, the selected patients (parents in the case of patients who were children) signed informed consent forms for

Table 1 Type of teeth and mean age of experimental groups

Groups	Anterior	Years	Posterior	Years
Conventional group (A)	1	14	4	30, 12, 39, 22
Laser-assisted group (B)	1	40	4	20, 22, 30, 27

Fig. 1 Preoperative radiograph

participation in the study. The study was done free charge for the control group patients, and it was registered at ClinicalTrials.gov, where it was identified as NCT01907919.

Experiment

Each group to be treated with the different techniques (group A, conventional technique; group B, laser-assisted technique) was identified; for each of these groups, five cases were chosen and treated with the same method under rubber dam isolation. Given the limited number of treated teeth and the fact that we were conducting a pilot study, the runs test was used to decide randomly whether a dataset was selected, and statistical analyses based on the descriptive method were conducted for selected datasets. All of the patients' teeth were treated by the same dentist. In both groups, if the bleeding continued for more than 3 min after exposure of pulp, the tooth was eliminated from the study. The efficacy of each technique as a therapeutic modality for preserving pulpal vitality was assessed by a recall program at 1, 6, and 12 months. The criteria used to evaluate the results of the different treatments were clinical and radiographic observations and findings, including the following:

1. Symptoms
2. Reactions to thermal stimuli, i.e., none, brief, and prolonged
3. Periapical radiographic changes
4. Radiographic-verified "bridge" formation

The occurrence of any one or more of the above symptoms in a patient resulted in the treatment's being classified as a failure.

Group A: five teeth were treated with traditional, rotating instruments; the cavities were prepared with diamond drills by turbine, multiple-blade drills that removed the decayed dentine, and, after cleaning the tooth and controlling the bleeding with cotton pellets and saline solution, resin modified glass ionomer cement (RM-GIC) (Vitrebond) (3M ESPE, St. Paul, MN, USA) light-hardening paste was placed gently on the exposed part of the pulp and the floor of the cavity. Then, the final restoration was completed using a layer of flowable Z350 XT and then P60 filled composites (3M ESPE, USA).

Group B: five cavities in the teeth were prepared in the same way as group A with traditional rotating instruments (Fig. 2); a hemostatic agent was used and the cavities were decontaminated by the diode 808-nm, laser-assisted procedure (Picasso-AMD, USA) (Fig. 3) using the following parameters:

1. Hemostatic agent: 1.5 W (Watt), continuous wave (CW), fiber diameter of 400 μm, in contact, 2 s per 1 mm, vertical and horizontal scanning movement on the exposure site (Fig. 3).
2. Decontamination of the cavity: 1 W, CW, fiber diameter of 400 μm, in contact, 2 mm per s, circular movement (Fig. 3). This capability achieves due to photothermal effect of diode laser via deep penetration and significant scattering through hydroxyapatite and water in the exposure's contaminated surroundings [28–30].

After application of the hemostatic and decontamination of the cavity, resin-modified glass ionomer cement (RM-GIC) (Vitrebond) (3M ESPE, St. Paul, MN, USA) light-hardening paste was placed on the exposed part of the pulp and the floor of the cavity (Fig. 4); this was followed by the final restoration with a layer of flowable Z350 XT and then P60 filled composites (3 M ESPE, USA).

In all of the patients, the restoration of the vitality of the teeth was assessed, apart from radiographic examination (Figs. 1 and 5), by conducting the following clinical tests:

1. Cold test by applying ice on the buccal side
2. Percussion test to forecast the periradicular involvement of the tooth
3. Palpation test to forecast the extent of periradicular bone involvement

Fig. 2 Pulp exposure during drill cavity preparation

Fig. 3 Pulp tissue coagulation/cavity decontamination with Diode laser

Fig. 5 One year postoperative radiograph

Results

Regarding the limited number of treated elements, the runs test was used to randomly decide if a set would be selected, and statistical analysis was based on descriptive data analysis. The collected data were entered into a computer and analyzed using the statistical software SPSS, version 17 (SPSS Inc., Chicago, Illinois, USA). The *P* value was greater than significance level (i.e., 0.05), which means that the hypothesis of being random was accepted. In group A, which included teeth treated with resin-modified glass ionomer cement (RM-GIC) (Vitrebond) (3M ESPE, St. Paul, MN, USA) light-hardening paste alone, two posterior teeth had not maintained their vitality after 2 and 6 months, while three of the five teeth maintained their vitality after 1 year (a total success rate of 60 %); reviewing literature do not include certain results as the clinical outcome of conventional pulp capping to the maintenance of the vital pulp; [18] Tronstad and Mjor [19] reported chance of success less than 50 % in caries exposure, while Al-Hyasat et al. [20] reported 33 % for caries exposure. In group B (those treated with the diode laser, 808-nm+resin-modified glass ionomer cement (RM-GIC) (Vitrebond) (3 M ESPE, St. Paul, MN, USA) light-hardening paste), after 1 year, all five teeth that were treated had maintained their vitality, indicating a success rate of 100 %.

A significant observation was made related to the ages of the failed cases in group A, i.e., they were the two youngest members of the study population, i.e., 12 and 22 years old. This reflects the importance of free microorganism than age in prediction of the pulp capping outcome in caries exposure (because success in pulp capping basically depends on the degree to which the wound area can be maintained free of bacteria). Teeth treated by the conventional technique showed slight sensitivity to heat stimulation, whereas, for patients treated with the diode laser, 808-nm+resin-modified glass ionomer cement (RM-GIC) (Vitrebond) (3 M ESPE, St. Paul, MN, USA) light-hardening paste, any sensitivity that resulted was due the bio-stimulation effect of the laser.

Fig. 4 Pulp tissue capping with resin-modified glass ionomer (Vitrebond)

Discussion

Pulp capping, which based on the healing ability of pulp tissue [5], is a subject of great controversy in dentistry. Pulp capping is used to treat accidental exposure of the pulp, traumatic lesions, and the removal of caries [2, 3]. Reviewing the pertinent literature showed high success rates for pulp capping in treating traumatic pulp exposure [6], but its effectiveness in treating carious exposures with bacterial contamination was low [7, 19, 20]. There are still controversies in this area among practitioners concerning the important decision of which treatment modalities are more effective in the contemporary practice of pulp exposure therapy. Successful pulp capping is obtained when the following clinical conditions are met, i.e., the pulp is not inflamed, adequate control of hemorrhaging, a good antibacterial seal exists, and the capping material is well tolerated by the pulp tissue [21]. The results of the previous studies indicated that, in both procedures (conventional and laser-supported), success basically depends on the degree to which the area of the wound can be maintained free of bacteria in the short term and in the long term. The end result should be the formation of a complete bridge of reparative dentin in direct contact with nearby reactionary dentin, as long as the tooth has acceptable physical and mechanical properties.

The results of this study were in agreement with those of various previous studies that concluded that laser-assisted

pulp capping has significant benefits in comparison with conventional procedures [19, 20, 22–27].

The principal advantages of laser-supported, direct pulp capping that produced better results in laser-treated group include the following:

1. Decontamination effect
2. Hemostatic effect
3. Bio-stimulation effect

Decontamination effect

Near infrared wavelengths, such as those provided by diode and Nd: YAG lasers penetrate deeply (500–1,100 μm) in dentin and have significant scattering capacity, whereas medium infrared devices, such as those in the CO_2 and erbium families of lasers, provide superficial penetration. These capacities (deep penetration and significant scattering) are extremely important in carious exposures when the exposure's surroundings are contaminated by different microorganisms [28–30]. Many authors have emphasized the importance of a sterile field as a crucial phase of the pulp capping treatment, and various studies confirmed the superiority of laser-assisted treatment over conventional, antibacterial agents.

Hemostatic effect

Hemorrhage control at the exposure site of the pulp is critical for the success of the pulp-capping treatment, especially when only an adhesive system is using as the capping material [31–35], as was the case in this study. The importance of a hemostatic for the exposed pulp is twofold, i.e., (1) increased bleeding can be indicative of a greater degree of inflammation in the pulp, resulting in diminished capacity for repair and (2) the moisture and contamination of the dentin adjacent to the exposure site due to bleeding can make it more difficult to obtain an adequate seal that will prevent subsequent bacterial exposure [36]. According to a study conducted by Swift et al. [21], various methods have been proposed to achieve pulpal hemostasis, including mechanical pressure with a sterile dry cotton pellet and various hemostatic agents, and each method has its limitations and inefficiencies [21]. The hemostatic efficacy of lasers, especially diode and Nd: YAG lasers, is due to the significant absorption of laser light by hemoglobin and melanin, which assures that the treated area will be dry in the least possible time. The hemostatic area is more superficial when $CO_2$and erbium family lasers are used, but the diode laser provides deeper penetration. The fine area of hemostasis created by the diode laser includes a thin layer of necrosis layer below which there is an area where the injury can be reversed, i.e., a place for the migration of inflammatory cells and the fibroblasts that contribute to the formation of the dentinal bridge.

Using laser-assisted, direct pulp capping, both irreversible and reversible reactions can be induced by the photothermal effect. The photothermal effect of laser light produces different morphological modifications in the surface of the pulp tissue, including vaporization, carbonization, homeostasis, and protein denaturation [37, 38]. In the direct pulp capping treatment, inadequate homeostasis can cause micro-leakage by hampering the bonding hybridization of the dentin substrate [36]. According to Hafez et al. [34] the application of 3 % NaOCl effectively promoted hemostasis, but renewed bleeding was sometimes evident at the margin of the pulp–dentin exposure after etching or priming of the adhesive system; by comparison, laser-assisted treatment provides assurance that the area will be dry.

Bio-stimulation effect

The bio-stimulation effect is a common characteristic of all lasers. According to Benedicenti [39], the use of a diode that has a wavelength in the range of 810 to 903 nm is especially advantageous in the healing processes of the pulp tissue [36]. The bio-stimulation effect of a diode laser provides reductions of inflammation and pain, cell proliferation and migration, cytodifferentiation of odontoblast-like cells, synthesis of dentin extracellular matrix, and the formation of reparative dentin in the injured pulp tissue [37, 38, 40].

In this study, the absence of any symptoms in the laser-assisted group was due to the benefits provided by the bio-stimulation effect of laser light. In the conventional group, the necrotizing action chemically induced by resin-modified glass ionomer cement (RM-GIC) (Vitrebond) (3M ESPE, St. Paul, MN, USA) light-hardening paste that do a part of inflammatory process of the pulp in direct contact with capping material, which could be a result of chemical irritation [41] and appeared that this necrotic layer is required for formation of reparative dentin [27, 42–45] whereas, in the laser group, the dentist had the ability to perform a more suitable procedure due to the physical action of laser light, the efficiency of which depended on understanding the laser-tissue interactions.[27]

Conclusions

It is clear that laser-supported pulp capping offers new treatment opportunities and enhanced working conditions, including the biological quality of the irradiated area and the increased efficiency of the interaction between pulp tissues and the capping material. The effectiveness of laser therapy has been demonstrated in direct pulp capping of teeth with carious exposures. There is only superficial contamination of the pulp tissue because of the capacity of the laser treatment to stimulate the formation of dentin by the pulp tissue, and the laser treatment also has the ability to decontaminate the irradiated

area and reduce the levels of pain sensed by the patient. Research on a larger sample, including posttreatment follow-up, should be conducted in future studies of this topic.

Acknowledgments This study was extracted from a thesis prepared by Iraj Yazdanfar in fulfillment of the requirements for the Master of Science degree in Laser Dentistry at the RWTH Aachen University's Hospital Department for Restorative and Conservative Dentistry and Periodontology in Aachen, Germany, where he had the honor and privilege of studying and working under the direction of Professor Norbert Gutknecht. I also offer my special thanks to Dr. Mehrdad Jalalian and Dr. Alireza Fallah for their guidance with the scientific writing and his support in the publication of this paper.

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